



HHS Health Sector AI RFI Response

Office of the Deputy Secretary and Assistant Secretary for Technology Policy (ASTP), Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC), Department of Health and Human Services.

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1.0 Executive Summary

This document provides a comprehensive examination of the challenges and opportunities associated with the integration of artificial intelligence (AI) into clinical care and outlines proactive measures that HHS can take to accelerate responsible and ethical adoption. AI has the potential to be transformative in modernizing healthcare systems, improving operational efficiencies, reducing clinician burnout, enhancing patient outcomes, improving diagnostic accuracy, and expanding access to underserved populations. However, any strategy to accelerate AI in clinical settings will need to address barriers to adoption, including fragmented data governance, low trust in AI systems, liability questions, demographic bias, misaligned incentives, and workflow friction. As a trusted partner in government transformation and digital enablement, LMI advocates for forward-thinking strategies that promote the benefits of AI adoption and scalability while maintaining a robust framework for safety, equity, and long-term sustainability.

Key Points:

Challenges to Adoption

- **Trust & Liability:** Clinicians and patients require confidence in AI accuracy and fairness, while liability ambiguities complicate adoption.
- **Outdated Infrastructure and Data Fragmentation:** Fragmented and biased healthcare data undermines the scalability and reliability of AI systems across diverse populations. Additionally, aging infrastructure in large healthcare systems often proves incompatible with modern AI platforms, limiting real-time data exchange and promoting the use of standalone AI tools instead of seamless integration into existing systems.
- **Operational Friction:** Lack of human-centered design and clinician/patient perspectives during development results in AI tools that fail to integrate seamlessly into workflows, leading to "friction-filled" systems rather than optimized productivity.
- **Economic Misalignment:** Outdated fee-for-service models do not incentivize value-based outcomes achievable through AI capabilities.
- **Administrative Barriers to Adoption:** Fragmented governance and oversight, lack of resources or understanding around change management processes, complex regulatory requirements, lack of standardized evaluation frameworks, and uncertainty around funding and reimbursement all contribute to delayed AI deployment.

AI in Healthcare: Success so Far

- **Reduced Administrative Burnout:** Research shows that AI scribes have resulted in decreased clinician burnout and increased patient capacity. A survey by the American Medical Association in 2024 found that the percentage of medical practices using AI almost doubled from 2023 and that clinicians are most comfortable with using these tools to assist with administrative tasks, rather than direct patient care.
- **Imaging and Diagnostics:** Several emerging AI tools for radiology and pathology demonstrate accuracy that rivals or exceeds human accuracy in detecting disease (Jones et al., 2026; Phan et al., 2025).

- **Operational Efficiency:** AI-driven scheduling has increased productivity and improved patient satisfaction by reducing wait times (Waligora et al, 2025).
- **Patient Triage and Virtual Assistance:** Chatbots and symptom-checkers in patient portals have been successful in reducing unnecessary emergency room visits and provide safe triage advice (Elechi et al., 2025).

LMI Recommendations for Strategic AI Integration

Recommendations for accelerating AI adoption span multiple modalities and are outlined below.

- **Regulatory Modernization:** To address barriers related to trust in AI, HHS can focus on the establishment of *risk-based oversight frameworks*, *standardized data privacy requirements*, and *algorithm transparency and performance documentation*.
- **Governance Frameworks:** Introduce guidelines for or creating standardized contracting templates, transparency protocols, and model-lifecycle controls to *address accountability gaps* in non-medical AI tools.
- **Private Sector Collaboration:** Foster public-private partnerships for scalable testing infrastructure and enable third-party certification frameworks to *encourage independent validation of AI solutions*.
- **Research and Development Priorities:** Focus on *explainable AI outputs*, *robust standards for data quality*, *performance benchmarks*, and *post-deployment monitoring* and drift detection to ensure AI systems evolve alongside dynamic healthcare practices without undermining reliability.
- **Surfacing Patient Perspectives:** Patient perspectives have not been adequately researched, but among those studies that have been done, the need for patient stakeholders in AI development, human oversight, transparency, and importance of data privacy have all been identified as key components for addressing patient trust in AI technologies.
- **Workflow Integration and Organizational Change Management:** Prioritize *human-in-the-loop models* to incorporate clinician oversight, usability optimization, and cognitive burden reduction for seamless adoption. Combine *usability testing*, *training strategies*, and *integrated change management* to drive trusted adoption and scalable tools.
- **Innovative Payment Models:** Transition from fee-for-service to *value-based care models*, like the model used by the CMS Advancing Chronic Care with Effective, Scalable Solutions (ACCESS) initiative, to create incentives for adopting AI systems that measurably improve patient outcomes and hospital efficiencies. Develop reimbursement pathways that align with AI's unique capacity to achieve long-term operational and clinical efficiencies.

AI holds immense promise for revolutionizing clinical care by addressing clinician burnout, improving diagnosis, and increasing access for underserved populations. However, realizing this potential requires coordinated regulatory reforms, better data governance, clear reimbursement frameworks, human-centered implementation strategies, and alignment among clinicians, patients, vendors, and healthcare systems. By addressing these barriers collectively, the U.S. healthcare system can leverage AI to deliver safe, equitable, and high-quality care at scale.

2.0 Defining AI

Before discussing strategies for promoting the development and adoption of AI tools it is important to understand what “AI” means. The term “Artificial Intelligence” or “AI” is frequently used inconsistently and interchangeably to describe different capabilities and use cases, particularly in vendor communications and popular discourse. This ambiguity complicates evaluation of actual system behavior, associated risks, and appropriate oversight.

For the purposes of this document, AI is defined as falling into one of the following two categories:

Machine Learning (ML)

ML techniques are already widely deployed in healthcare and include both traditional statistical learning and modern deep learning methods. These systems typically operate on structured or semi-structured data and produce bounded, task-specific outputs.

Examples include:

- Risk prediction models (e.g., sepsis risk, readmission risk)
- Medical imaging and signal analysis (e.g., radiology, pathology, ECG interpretation)
- Population health stratification and resource optimization
- Clinical decision support rules enhanced with learned parameters

These systems are often deterministic at inference time, evaluated against predefined performance metrics, and more amenable to validation within existing regulatory and quality frameworks.

Foundation models and Large Language Models (LLMs)

Foundation models, such as LLMs and multimodal models, are pre-trained on large, heterogeneous datasets and adapted for downstream tasks and generation of ‘new’ content.

Examples include:

1. Clinical documentation assistance and summarization
2. Natural language interfaces to EHR data
3. Patient-facing chatbots and triage assistants
4. Multimodal models combining text, images, and signals

Unlike many traditional ML systems, these models:

- Generate open-ended outputs
- Exhibit probabilistic and context-sensitive behavior
- Produce fluent but sometimes incorrect or misleading responses
- Introduce new risks such as hallucination, automation bias, cognitive load, and alert fatigue

As a result, LLMs and other foundation models require distinct evaluation methods, workflow testing, and human-centered safeguards. In the document below, LMI will speak to both categories

of AI, their application in clinical spaces, and the ways HHS might support, promote, and regulate their use.

3.0 Regulation and Adoption

Existing Barriers to AI Adoption in Clinical Settings:

With a solid foundation in the scope and capabilities of AI, the focus now shifts to the critical challenges and opportunities that influence its integration into clinical care. Private sector innovation in AI for health care and its adoption in clinical care is shaped by a range of barriers that differ depending on the stage of development, implementation, and the stakeholders involved. These include:

Trust in AI: Clinicians and patients alike must have confidence in the safety, accuracy, and overall performance of AI tools. Clinicians remain liable for patient care, meaning AI must meet high standards of reliability, while patient trust depends on accurate, unbiased tools for decision-making about their care.

Liability concerns: Both clinicians and AI vendors face risks related to clinical errors caused by AI systems. While clinicians are legally accountable for patient outcomes, AI vendors are increasingly being called upon to share liability for faulty algorithms and design defects.

AI System Biases: AI systems may unintentionally prioritize certain variables or identify invalid correlations, leading to biased outcomes. The lack of transparency in decision-making ("black box" issues) further complicates efforts to address these biases. AI tools are often more accessible in higher-resource settings, potentially widening healthcare gaps for underserved populations. Over-reliance on AI without human oversight may overlook critical individual and contextual factors as well. Mitigating such biases requires diverse datasets, regular auditing, human oversight, and inclusive collaboration to ensure these technologies improve care for all populations equitably (James, 2024). There are several examples of biased AI models resulting in discrepancies in patient care and outcomes. An independent study found that black patients were not identified with respiratory illness due to pulse oximeters being less accurate on darker skin. (UK Department of Health and Social Care, 2024) In a review of an algorithmic model to identify liver disease, AI missed it in 44% of female patients vs 23% of males. (Straw & Wu, 2022).

Data Access: Access to high-quality, diverse, non-fragmented data is critical for developing robust AI models. Health data is often siloed across multiple systems and formats, making it difficult to aggregate and standardize. Also, privacy and security constraints such as HIPAA and PHI make data sharing and reuse complex. Even when data can be accessed, concerns over breaches, confidentiality and patient consent slow down innovation.

Legacy Systems and Interoperability: Large healthcare systems often rely on aging infrastructure that cannot easily communicate with modern AI platforms. Incompatible data systems and lack of standardized data formats (e.g., FHIR) hinder real-time data exchange. In addition, many AI tools are implemented as separate, "bolted-on" applications rather than being seamlessly integrated into systems such as the electronic health record. This forces clinicians and

care teams to toggle between systems causing frustration and workarounds which impact workflows, safety and efficiency (Pavuluri et al., 2024).

Disruption of Established Workflows: Poorly integrated AI can increase the cognitive load on clinicians and care teams rather than reducing it. If an AI tool adds extra steps to already busy, high-stress workflows for clinicians and care teams, it is likely to be abandoned (Elechi et al., 2025).

Analysis of the Barriers to AI Adoption in Healthcare

Early machine learning-based AI applications have exhibited model drift, algorithmic errors, unintended biases, and overfitting, leading to decreased accuracy, classification or prediction errors, and reduced ability to generalize. Generative AI applications can experience all these problems as well as “hallucinations,” and “sycophantic behavior” leading to blatantly incorrect or difficult-to-detect false information that can harm patients (Christophe et al., 2026; Peng et al., 2026; Sittig & Singh, 2025).

Together, these barriers form a multi-layered ecosystem problem — not just a technical one. Overcoming them requires coordinated action across:

- Policy and regulations to clarify standards and liability
- Data governance that balances privacy with ethical access
- Clinical partnerships to build trusted evidence and workflow fit
- Funding and incentives to support innovation and long-term ROI
- Education and change management to build clinician confidence

Only by addressing these factors in tandem can private-sector AI innovation truly translate into reliable, safe, and widely used tools in clinical care.

While tools incorporating AI are rapidly making their way into clinical practice, mechanisms to ensure their **safety** and **effectiveness** remain underdeveloped. For instance, there is little guidance for health care organizations (HCOs) and clinicians as they navigate the complexities of implementing AI into routine care. To ensure that integrating AI-enabled systems into health care **transforms clinical decision-making** and **improves patient outcomes**, it is essential to proactively address known risks and establish safeguards to manage potential unknown risks as they emerge.

Clinicians are still liable for patient care quality and outcomes and therefore need to feel that the AI input is reliable and accurate. For example, documentation tools such as Deep Scribe or Freed reduce administrative workload and empower providers to focus on patient care by automating transcription. However, the Scribe doesn’t always get the transcription 100% right and human involvement is needed to ensure accuracy and maintain patient safety. To integrate AI safely, there is an ongoing need for a “human-in-the-loop”. This applies not only to administrative clinical applications, but also to diagnostic applications of AI. Algorithms have shown proficiency in detecting conditions such as tumors, malaria, and prenatal issues, performing at or above expert levels in certain cases (Fan et al., 2025; Faratisha et al., 2026; Isewon, Oba, & Oyelade, 2025; Kumar et al., 2025; Nishida et al., 2022; Pai et al., 2025; Yeganegi et al., 2025). However,

diagnostic AI has also exhibited biases, for example, relying on factors like medical spending as proxies for care needs, which can lead to mislabeling or overlooking vulnerable patients (Obermeyer et al., 2019).

Despite these challenges, AI offers hope for improving health care efficiency and accessibility, especially amid shortages of clinicians and nurses. Healthcare systems often operate at razor-thin margins, and AI has the potential to reduce personnel burdens, increase accuracy, and expand access, particularly for underserved rural populations. AI tools that address clinician burnout—such as enhancing documentation or decision support—are already easing operational strain and enabling quality time with patients (Albrecht et al., 2024; Shah et al., 2025).

Concerns around liability where both the clinician (for overreliance on AI) and the AI vendor (for defective design) can both be liable also represent a barrier to adoption. There are also initiatives to hold AI vendors responsible for clinical harm. Previously, once the sale of AI capabilities was complete, all risk transferred to the provider. More recently, there have been initiatives to also hold AI vendors responsible for clinical harm. AI vendors are now being asked to not only provide capabilities but to also solve problems and to share risk. Clarifying the boundaries of liability is key for encouraging adoption and integration of AI in clinical settings, while also ensuring product vendors are providing safe and reliable tools.

AI has been available since 2007 and as its capabilities grow, clinician trust continues to grow as well. With 27 million uninsured Americans, over 100 million lacking a primary care provider, and 200,000 annual deaths each year from preventable medical errors, AI offers a way to ease documentation burdens, reduce provider burnout, and improve focus on patient care (Briones & Cohen, 2026; NACHC, 2023; Rodziewicz, et al., 2025). AI holds promise for enhancing efficiency, cutting costs, and expanding access, particularly in underserved rural areas. In an overburdened healthcare system with significant shortage of nurses and physicians AI presents opportunities for trusted scalable care.

LMI Policy Recommendations for Addressing Barriers to AI Adoption:

As HHS aims to shift the U.S. healthcare system toward "deflationary technologies"—tools that improve health outcomes while simultaneously lowering long-term costs, the current regulatory, payment policy and program design create an environment where AI adoption may remain limited to "experimentation" rather than becoming a standard part of safe, high-quality clinical care.

To incentivize AI in clinical care, HHS should focus on modernizing outdated systems to ensure they do not hinder innovation. **Key priorities involve moving from rigid fee-for-service models to outcomes-based incentives and establishing clear, risk-based regulatory guardrails.**

Revising current Payment Policy to reflect a prioritization of efforts and programs aimed at transitioning from episodic or fee-for-service payment models to alternative models that incentivize and reward the long-term efficiencies possible through reasonable adoption of AI technologies is one part of a strategy that will incentivize the adoption of AI in healthcare. Prioritizing emerging models like the CMS ACCESS Model (CMS, 2025) which aims to provide stable, recurring payments for technology supported chronic care rather than traditional fee-for-service payment models is a start on this journey. Coupling the CMS ACCESS Model program

with efforts that incentive providers who adopt AI integrations that demonstrably reduce hospitalizations and/or improve diagnostic accuracy through “bonus points” or increased shared savings opportunity are potential payment policy changes that would encourage the reasonable AI adoption sought after here.

When examining the regulatory landscape, opportunities exist to remove the “bottlenecks” that currently exist for developers and healthcare systems alike. Proposed regulatory changes to remove these barriers and incentivize the effective use of AI in healthcare include a transition to a Risk-Based Oversight model that employs rigidity on a sliding scale that is proportionate to the risk itself. For example, low risk tools (e.g. administrative automation) would be subject to less rigorous regulatory hurdles while high risk/impact tools (e.g. appeals and grievances and utilization management) require increased scrutiny and a higher regulatory bar to clear.

Other regulatory design changes to consider would include:

1. Addressing **Health IT Certification and Transparency** by updating the **ONC Health IT Certification Program** to include criteria for algorithm transparency and decision support, ensuring clinicians understand and trust AI-generated recommendations. Specifically, implement and refine the **Health Data, Technology, and Interoperability (HTI-1)** requirements under **45 CFR Part 170** to strengthen decision support transparency that requires certified health IT to provide users with a "consistent, baseline set of information" about algorithms, including their fairness, validity, and safety as well as simplifying the certification process for AI models to reduce development costs and administrative burdens.
2. **Data Privacy and Data De-identification:** Modernize **45 CFR § 164.514(b)** to address the risks of re-identification in the age of big data and AI. Specifically, move beyond the static “safe harbor” list of 18 identifiers to as few as four pieces of non-exclusive data can now re-identify an individual. Updates should provide more flexible, risk-based de-identification methods that allow for richer datasets while maintaining high privacy standards.
3. **Medicare Advantage and Prior Authorization:** Update **42 CFR § 422.101(b)(6)** and **42 CFR § 422.138** regarding utilization management and prior authorization to explicitly state that Medicare Advantage (MA) Plans cannot use AI alone to deny coverage; decisions must be based on an individual’s specific circumstance and a must retain a Human-in-the-Loop. MA plans must make publicly available any internal coverage criteria used by the AI models.
4. **FDA Lifecycle Management for AI Devices:** Shift from “static” device approvals to dynamic lifecycle oversight by formalizing the Predetermined Change Control Plans (PCCP) within the existing frameworks of **21 CFR Part 820** (Quality System Regulation) and marketing pathways (510(k), De Novo, PMA) to allow a PCCP to pre-authorize specific future algorithm updates (e.g., iterative machine learning improvements) without needing a new marketing submission for every change, provided they maintain rigorous quality standards.

Other programmatic design changes to support AI adoption include:

1. **Create AI Implementation Grants** through the CMS Innovation Center (CMMI) to pilot "AI Accelerators" that provide upfront technical assistance grants for training and workflow integration.
2. **Establish Interoperability Standards** that ensure “data liquidity” is maintained for AI tools to pull/push information across platforms with minimal technical barriers.

3. **Clarify of Liability Guidelines** to protect clinicians when AI models produce incorrect or harmful recommendations. Currently, the risk of legal exposure in instances where AI makes an error would be a major barrier to adoption of AI technologies in a clinical setting.

Administrative Influence on AI Adoption

Updated policy and regulation cannot drive AI integration in a vacuum. AI adoption in clinical care is shaped by coordinated governance across clinical, technical, operational, and compliance stakeholders. Administrative barriers, including fragmented oversight, procurement complexity, regulatory requirements, and organizational change management, frequently present greater challenges than technical limitations. Addressing these barriers through standardized governance models, clear evaluation processes, and early multidisciplinary engagement will be critical to advancing responsible and scalable AI adoption across healthcare organizations.

Adoption of new technology and processes is governed by a multidisciplinary decision structure that balances patient safety, regulatory compliance, clinical effectiveness, operational feasibility, and responsible data stewardship. Healthcare organizations typically require coordinated approval across executive leadership, clinical governance bodies, technical authorities, compliance functions, and operational stakeholders. Understanding these roles and the administrative barriers they face is critical to enabling responsible, scalable AI adoption aligned with HHS priorities.

Key Decision Makers and Governing Bodies

- **Clinical and Executive Leadership**—including Chief Medical Officers (CMOs), Chief Clinical or Nursing Officers (CCOs/CNOs), and executive leadership teams—determine whether AI solutions align with clinical quality goals, workforce needs, patient safety standards, and organizational risk tolerance. Clinical leadership endorsement is often the most significant driver of institutional trust and adoption readiness.
- **Information Technology, Data, and Security Leadership**—including Chief Information Officers (CIOs), Chief Digital or Analytics Officers, and Chief Information Security Officers (CISOs)—serve as technical gatekeepers responsible for evaluating interoperability with electronic health records, cybersecurity risks, infrastructure readiness, and data governance compliance. Their approval is essential to implementation feasibility.
- **Clinical Governance and Patient Safety Bodies**—such as Clinical Informatics Committees, Quality and Patient Safety Boards inclusive of patient advisory councils, Clinical Decision Support governance groups, and Institutional Review Boards (IRBs)—assess clinical evidence, bias mitigation strategies, monitoring requirements, and overall patient risk. These bodies determine whether AI solutions meet standards for safe and equitable clinical use.
- **Finance, Compliance, and Acquisition Functions**—including Chief Financial Officers (CFOs), acquisition offices, legal counsel, and compliance teams—evaluate procurement requirements, cost sustainability, reimbursement considerations, vendor risk, and adherence to federal privacy and security regulations.
- **Operational and Frontline Clinical Leadership**—including physician champions, nursing leaders, department chiefs, and clinical informaticists—validate workflow

integration, support clinician adoption, and ensure solutions align with real-world care delivery practices.

Primary Administrative Barriers to Adoption

Healthcare organizations face several administrative challenges that can delay AI deployment:

- **Fragmented Governance and Oversight:** Multiple committees and unclear approval pathways can create lengthy evaluation cycles, particularly in organizations without formal AI governance frameworks.
- **Complex Regulatory and Compliance Requirements:** Privacy protections, safety oversight, liability considerations, and evolving regulatory expectations increase review timelines prior to clinical use.
- **Acquisition and Contracting Complexity:** AI solutions introduce nontraditional licensing models, data use agreements, and cybersecurity assessments that extend procurement processes.
- **Lack of Standardized Evaluation Frameworks:** Organizations often lack consistent methodologies for assessing clinical effectiveness, operational impact, and algorithm performance across diverse patient populations.
- **Operational and Change Management Demands:** Implementation requires updates to clinical workflows, policies, training programs, and documentation practices, which can slow adoption without coordinated planning.
- **Funding and Reimbursement Uncertainty:** Misalignment between budgets, uncertain reimbursement pathways, and competing organizational priorities can delay investment decisions.

4.0 Reimbursement: Feedback on how HHS can approach reimbursement regulation to drive AI adoption

One of the primary barriers to AI adoption and development is the uncertainty around funding and reimbursement. To align payment policies with the goal of promoting high-value clinical AI, fostering competition, and increasing affordability, the following key policy changes should be considered:

- **Establishing Predictable Reimbursement Pathways** through proposals like Health Tech Investment Act (S. 1399) which would require CMS to place qualifying AI services into a "New Technology" Ambulatory Payment Classification (APC) for at least five years and/or **Transitional Add-on Payments**, similar to the Transitional Drug Add-on Payment (TDAP), which provide temporary extra funding for new AI tools, allowing CMS to gather data on their value before incorporating them into permanent bundled payments ("Medicare reimbursement pathway for AI-enabled medical devices considered in Senate's Health Tech Investment Act", 2025; Levins, 2022).
- **Prevention of the "overuse" of risks associated with fee-for-service models** by shifting toward Value-Based and Outcome-Aligned Models such as the ACCESS Model and continuing to encourage contracts where manufacturers share the financial risk. For example, providers might receive rebates if an AI tool fails to deliver the promised clinical or economic outcome.

- **Incentivizing Equity, Interoperability, and Competition** via policy changes designed to ensure AI tools work for diverse populations and aren't restricted to wealthy, urban health systems. Proposed changes include financial rewards for **bias mitigation** where payers may offer higher reimbursement rates for AI tools that demonstrate **interoperability** and proven effectiveness across multiple demographic groups; and premarket Advance Market Commitments (AMCs) that offer monetary prizes for solving specific clinical challenges, with the requirement that the code eventually becomes more accessible to spur follow-on competition.
- **Modernizing Medicare Advantage (MA) Oversight** with guidance to ensure AI tools are used to support, not supplant clinical judgment. Guidance should include a prohibition on automation bias meaning insurers cannot use AI to override established medical necessity standards or discriminate against specific patient histories, ensuring that high-value care remains the priority over automated cost-cutting.

Analysis of Reimbursement Challenges Facing AI Integration in Clinical Spaces

Many AI innovations improve or enhance workflows through workflow infrastructure (e.g. automate components of a workflow) or as decision support (e.g. make prescriptive recommendations). However, these capabilities don't fit neatly into legacy health care billing structures, which are traditionally designed around billable procedures and specific outcomes. For example, an AI tool which assists in diagnosis or notes summarization may improve decision support or save time, but that's not a specific procedure or billable outcome. This misalignment raises critical questions about reimbursement for AI-assisted or AI-enabled clinical functions. Many AI tools provide indirect benefits, such as improving provider workflows or enhancing decision-making, yet these contributions often involve broader infrastructure rather than discrete, billable outcomes.

To address these complexities, health care leaders need to define clearer reimbursement frameworks. Key considerations include establishing **what aspects of AI are reimbursable**—such as AI as a service, AI-powered infrastructure, or AI-enabled outputs—while ensuring clarity around safeguards like **human oversight, documentation, and quality control**. Additionally, it's essential to outline standard criteria to justify reimbursement, which may involve demonstrating **measurable clinical value, operational efficiency, or patient safety improvements**. Developing a comprehensive checklist of dimensions for assessing AI reimbursement capabilities will be crucial as the industry seeks to navigate this emerging area.

Payment models must evolve to reward measurable value rather than relying on traditional per-use reimbursement structures. Value in health care should be defined by outcomes such as improved patient care, enhanced clinical safety, or reduced administrative burden. For example, if an AI-enabled tool significantly decreases diagnostic errors or reduces administrative workload, these outcomes represent quantifiable improvements that could form the basis for reimbursement. By incentivizing outcomes over isolated usage, payment models can more effectively align with the capabilities and benefits of AI innovation. As AI continues to expand its role in health care, these efforts will ensure its benefits are realized in a financially sustainable and equitable manner.

5.1 Research and Development: Recommendations for prioritization of AI research and development subject areas

In addition to targeting policy changes for regulation, reimbursement, and oversight of AI adoption and integration into clinical workflows, it is equally important to invest in strategic innovation to address technical, operational, and ethical barriers. By aligning research priorities with the needs of health care providers and patients, organizations can develop AI solutions that not only meet policy requirements but also enhance practical usability and outcomes, paving the way for broader and sustainable integration into health care systems.

There are several key areas of AI research that should be prioritized to support and accelerate the adoption of AI in clinical care. These focus areas have been identified not only for their potential to drive adoption, but also for their ability to strengthen trust, reduce uncertainty, and implement guardrails for the responsible use of AI tools. These priorities are discussed below.

Criteria for AI prioritization. In addition to identifying application areas, it is crucial to define criteria that guide AI prioritization. These criteria ensure that research and development efforts align with core objectives for successful adoption. They include:

- **Adoption:** Facilitating widespread, rapid integration into clinical workflows with minimal disruptions.
- **Trust:** Building confidence among clinicians, patients, and the public in the safety and reliability and resiliency of AI solutions.
- **Reducing uncertainty:** Mitigating ambiguity around AI risks, outcomes, and implementation challenges, including administrative and reimbursement questions.
- **Guardrails:** Establishing clear guidelines and safeguards to ensure ethical and responsible use of AI technologies.

Research Target #1: Explainability for Accountability

HHS should prioritize research on AI systems that support clinical reasoning by making their outputs interpretable and aligned with how clinicians make decisions in real-world settings. **Care providers are ultimately accountable for patient outcomes and must be able to understand, validate and challenge AI generated care recommendations.** AI systems are at risk of conveying conclusions or diagnosis while not stating the confidence or communicating uncertainty. AI should be able to clearly identify which patient features or clinical events drive a recommendation in a way that is rooted in science and that the care providers can validate.

A multidisciplinary approach is essential in building healthcare AI to ensure safety, efficacy, and ethical compliance, as AI models require the combined expertise of data scientists, clinicians, patients, ethicists, and IT professionals to address complex patient data, regulatory requirements, and technical challenges. HHS should ensure such collaboration is supported to prevent algorithmic bias, build trust in clinical decisions, integrate AI into existing workflows, and monitor, audit, and update AI models as medical knowledge advances (Quinn et al., 2020).

Further, HHS should support research into how accountability is shared and documented across stakeholders and multidisciplinary teams (such as stakeholders, facilities, technical and medical

teams, etc.). If HHS can create an environment where care providers and patients can trust the AI recommendations and diagnosis, this will directly translate to increased adoption of AI as part of clinical care.

Research Target #2: Knowledge Management Capabilities

All data-driven approaches—from machine learning to generative AI to rules-based approaches and retrieval—depend on what content is accessible to the model, how that content is captured (e.g. database schema, unstructured document, etc.), and how it is stored. The success of any AI model is directly tied to the quality and structure of the data it relies upon. Fragmented data pipelines, such as siloed data sources, or unstructured, messy data, severely impair the functionality and reliability of any AI model. Addressing these challenges is crucial to unlocking the full potential of AI in clinical care. To this end, HHS should prioritize several key research areas in knowledge management:

1. **Digitalization, integration and structuring of content:** One of the foundational priorities for the effective adoption of AI in clinical care is the digitization and integration of health care data into a structured format. Healthcare data exists in a myriad of formats, each with varying levels of security and sensitivity. A focus on developing and implementing systems that securely integrate health care content while adhering to rigorous security standards is vital for building robust and reliable AI technology. This requires advancements in encryption, regulatory safeguards, and collaboration between technology developers and regulatory bodies. Work has been done in areas such as data security protocols, but further efforts are needed to ensure sensitive health care data can be effectively stored and processed while meeting privacy and compliance requirements.
2. **Safe use of imperfect health care data:** In real-world settings, health care data is often incomplete, noisy, or reflective of biases, as seen in many EHR systems. While perfect data may not always be attainable, HHS can prioritize research focused on the development of methods to use imperfect data safely and effectively. Techniques such as data quality assessments, validation protocols, and error detection frameworks can help mitigate risks associated with unreliable or inconsistent datasets. Furthermore, AI systems can be designed to operate with partial or fragmented information, with safeguards to prevent misinformation or harmful conclusions.
3. **Development of performance benchmarks and standards:** Trust is one of the most significant barriers to AI adoption in clinical care, and a lack of transparency regarding how models perform only exacerbates uncertainty among users. Establishing clear performance benchmarks and standards can address this issue by providing proof of performance that users—whether clinicians, patients, or regulators—can rely upon. By supporting research to develop standardized methods for measuring and reporting AI performance, HHS can ensure consistent evaluation practices across AI solutions.

Research Target #3: Improve AI Integration across Provider Network

Research focused on **human-centered coordination workflows**—particularly how AI can enhance shared situational awareness among clinicians, nurses, and care managers across a

provider network without increasing administrative burden—can help advance AI technology for seamless operations across organizations, data systems, and care teams. This makes it an especially relevant area for HHS leadership and investment. **By centering research on understanding how AI can fit into real clinical workflows and reduce fragmentation instead of adding extra noise into an already chaotic system, HHS can meet clinical staff, administrators, and patients where they are and address real world needs.**

As part of this focus, research should explore how AI outputs can be optimized for the unique roles of care team members. For example, clinicians, nurses, and office staff each require different levels of detail, patient information, and timing to perform their roles effectively. A well-designed AI system would account for these variations and contribute to improving patient care while minimizing alert fatigue, a critical factor for driving adoption of AI in clinical workflows. By tailoring AI tools to the specific needs of each care team member, this research can ensure that AI enhances care delivery across the network without adding unnecessary complexity.

A natural second phase of provider integration is the **post-deployment modeling of AI systems** integrated into health care systems. Key focus areas include model or data drift and updates to clinical practice patterns. Many AI tools demonstrate strong performance in controlled development or pilot settings. Far less is known about the system performance post deployment. As health care is always evolving, the AI used to support will need to keep pace as well. HHS should invest in research aimed at **monitoring key metrics such as diagnostic accuracy, clinician workload, and AI response time** and have the performance reports delivered in a way that does not impose additional documentation burden on clinicians. If care providers can have confidence that the AI they are working with is up to date, it will increase not only the adoption of and trust in the AI tool but also reduce the churn of AI tools in the future.

5.2 Review of Existing Literature on AI in Healthcare Settings

To best understand the most promising areas of Healthcare AI research and development, it is important to review the academic landscape as it currently exists. A growing body of published literature, including systematic reviews, meta-analyses, and observational studies, has examined how adopted AI tools influence clinical care, showcasing both potential benefits and emerging risks.

One significant area of research into the benefits of AI in clinical spaces has surfaced evidence showing that AI tools can achieve diagnostic performance comparable to human clinicians across multiple medical domains. Takita et al. (2025) conducted a systematic review and meta-analysis comparing generative AI with physicians, finding that AI models performed similarly to clinicians overall, though performance was slightly lower than expert-level physicians in some tasks. Complementing this, McGenity et al. (2024) reviewed AI applications in digital pathology and reported high diagnostic accuracy, with pooled sensitivity and specificity exceeding 90% across diverse histopathological tasks. A growing collection of studies indicate that AI can serve as a reliable adjunct to clinical judgment, supporting diagnostic decision-making while emphasizing the need for continued validation and careful integration. Beyond diagnostic accuracy, studies have also examined whether AI integration translates into tangible improvements in patient outcomes. Burdick et al. (2020) evaluated a sepsis prediction algorithm across multiple U.S. hospitals and found that its use was associated with reductions in in-hospital mortality, length of stay, and 30-

day readmissions. Similarly, Yuan et al. (2025) conducted a meta-analysis of AI-powered early warning systems and reported significant improvements in patient outcomes, including reduced mortality and timely recognition of clinical deterioration. Collectively, these findings suggest that **AI can enhance patient safety and care quality**, particularly when used to support early intervention and proactive clinical decision-making.

Evidence from systematic reviews suggests that adopted AI tools can positively influence clinical decision-making and care delivery across multiple healthcare roles as well. AI-enabled clinical decision support systems have been shown to improve diagnostic accuracy, treatment selection, medication safety, and workflow efficiency by providing patient-specific recommendations to clinicians (Aggarwal et al., 2024). Similarly, nursing-focused research indicates that AI technologies support earlier detection of patient deterioration, reduce hospital readmissions, improve documentation accuracy, and enhance nurses' confidence in clinical judgment (Mikkonen et al., 2025). However, both reviews highlight variability in implementation and **emphasize the need for standardized evaluation, ethical oversight, transparency, and adequate training** to ensure AI tools safely augment clinical practice. A systematic review and meta-analysis examining AI use in clinical documentation found that AI tools—particularly those using natural language processing and generative models—were associated with significant reductions in documentation time and workload (Zhao et al., 2026). AI-generated documentation was generally comparable in quality to clinician-authored notes, suggesting potential benefits for reducing clinician burnout. Nevertheless, the authors stressed that human oversight remains essential to maintain accuracy, safety, and accountability in clinical records.

Emerging evidence also highlights potential risks associated with routine AI assistance. For example, an observational study of colonoscopy practice reported that clinicians who routinely relied on AI support demonstrated reduced detection performance when the AI was removed, suggesting possible skill degradation over time (Budzyń et al., 2025). While causality cannot be definitively established, these findings raise concerns about over-reliance on AI and the importance of maintaining clinician expertise alongside technological augmentation. It is important to note that the lag between conducting research and publications often spans 1-2 years or more. In contrast the advancement in AI technologies is occurring quickly; as Dr. Dashun Wang puts it, “The rapid rise of generative AI—which is reshaping research workflows faster than many scientific institutions can keep up—only makes the stakes higher and the future shape of science less certain” (Zhau, 2026). The strategy HHS adopts to support adoption of AI will need to include consideration for the vast differences between the speed of research and the speed of innovation.

Economic Evaluation

The literature on artificial intelligence in clinical care frames the costs and benefits of AI adoption through a spectrum of economic evaluation methods, ranging from traditional cost-effectiveness analyses (CEA) and cost-utility analyses (CUA) to budget impact analyses (BIA) (El Arab & Al Moosa, 2025). Currently, there are **very few high-quality economic evaluations of AI in healthcare** (Wolff et al., 2020). Many studies report partial or isolated cost savings, and many omit key costs such as implementation and maintenance.

El Arab & Al Moosa, 2025 outline that many evaluations rely on static decision models that may overestimate benefits, as they often exclude long-term, dynamic learning effects and critical cost components such as infrastructure and training. In general, they found that across oncology, cardiology, ophthalmology, and infectious disease applications, **AI enhances diagnostic accuracy, increases quality-adjusted life years (QALYs), and reduces overall costs** by minimizing unnecessary procedures and optimizing resource allocation.

Complementing these findings, other systematic reviews highlight methodological gaps that temper overly optimistic conclusions. A separate review of health economic evaluations of medical AI tools found that while most studies report cost savings or cost effectiveness, reporting quality is often suboptimal and key cost categories are frequently omitted (Junior et al., 2026). For instance, only a minority of evaluations explicitly included implementation or operational costs, and many assessments focused on early-stage AI technologies with limited real-world validation. This raises concerns that published economic benefits may be inflated or insufficiently robust for policy or reimbursement decisions. The review calls for future studies to report complete cost components, adhere to rigorous economic reporting standards, and assess AI systems at higher technological maturity levels to ensure that cost benefits reflect real clinical deployment rather than theoretical or pilot settings.

Another blind spot in AI adoption cost reporting pertains to the economic evaluation of Software as a Medical Device (SaMD). While Wu et al. (2025) found that cost-utility and cost-effectiveness analyses across 52 studies generally reported AI interventions as cost-effective, cost-beneficial, or cost-saving, comparisons were difficult when studies excluded key expenses such as software licensing, maintenance fees, and app access costs from reporting. This inconsistency limits the reliability of conclusions about AI's true economic value and underscores the need for standardized frameworks that capture all relevant costs associated with SaMD.

Recent literature also emphasizes equity, bias, and distributional concerns. It was found that evaluations often fail to stratify results by demographic or socioeconomic subgroups, potentially masking disparities in who benefits from AI and who may incur costs or harms (El Arab & Al Moosa, 2025). Equity considerations such as differential AI performance across populations, varied access to digital infrastructure, or unequal capacity to absorb training and integration costs are underreported, highlighting the need for inclusive and context-sensitive economic frameworks that consider both monetary and justice-related dimensions of AI adoption.

In summary, current literature approaches the costs and benefits of AI in clinical care through a plurality of economic evaluation methods, with most evidence pointing to potential cost savings and improved clinical value, especially when AI enhances diagnostic accuracy or streamlines care delivery (Wolff et al., 2020; Wu et al., 2025). However, persistent methodological limitations (e.g., incomplete cost inclusion, static modeling), the early maturity of many technologies, and underrepresentation of equity impacts, caution against simple cost-benefit conclusions. Robust, transparent, and context-specific studies, especially those incorporating full cost components over time and subgroup analyses, are essential for translating promising economic signals into reliable guidance for policymakers, health systems, and payers (El Arab & Al Moosa, 2025).

6.1 AI Approach: Recommendations on how to support AI integration and responsible use in clinical settings

AI Evaluation and Oversight

The practical implementation of artificial intelligence (AI) tools in clinical care requires robust evaluation methods to ensure effectiveness, reliability, and human-centered integration. Pre-deployment assessments as well as post-deployment monitoring are both vital to ensuring AI systems are trustworthy from the moment they enter the clinical workflow and continue to be over time. Not only should evaluation focus on model function and accuracy, but it should also include human centered metrics for usability, integration into workflows, and training. This multifaceted approach paves the way for equitable and sustainable AI implementation across diverse healthcare settings.

In pursuit of AI adoption in clinical care, leveraging standardized AI evaluation benchmark tests—such as BIG-bench, Massive Multitask Language Understanding, GLUE and the Holistic Evaluation of Language Models—serves as a powerful pre-deployment assessment strategy. These tests provide valuable insights into the general reasoning, linguistic competence, and failure cases of AI models and provide visibility into how the models follow instructions, conduct multi-step reasoning, and ensure robustness across task types. AI models can be compared against each other and human performance to identify which ones are worth investing in for specific use cases where defined criteria are highly prioritized. A complete evaluation using these benchmark tests will allow evaluators to assess where the model performs well and where performance is inadequate. Pre-deployment robustness testing should be a high priority for HHS support as part of regulatory policy. This includes perturbation testing and sensitivity analysis to minor data changes or missing data. Models should be able to surface uncertainty and work with humans to review responses when confidence is low. Before deployment, shadow runs and simulated deployment testing can be very valuable in generating buy in with care providers without influencing care. Shadow runs allow assessment of model decisions, output timing and error patterns, enabling gaps to be identified and corrected before deployment.

Once deployed, AI systems must be continuously monitored to ensure that performance remains stable as patient populations, medical knowledge, care practices and data evolve. Support for evaluation approaches that enable early detection of model degradation and ensure timely updates before patient care is compromised is important for ensuring long term success of AI tools. One effective method to track post-deployment performance is input and output drift monitoring. Input drift tracks shifts in patient demographics, diagnosis prevalence care provider patterns, and documentation styles. This can be done using distributional distance metrics such as KL divergence and maximum mean discrepancy. Distributional distance metrics measure how different today's data look compared to the data when the model was first deployed, providing early signals when changes in patients, documentation, or workflows may threaten model reliability—even before errors are visible in clinical outcomes. Output drift monitoring tracks sudden changes such as an increase in high-risk flags or shift in recommendation thresholds. Furthermore, humans should be constantly “in the loop” to ensure that the output drift reflects real changes in care and not hallucinogenic spikes.

The literature on AI evaluation methods for non-medical devices demonstrates a clear trend: technical performance metrics are often prioritized, while workflow integration and human-centered evaluation approaches receive comparatively less attention. To assess the overall efficacy of an AI tool in clinical settings, researchers recommend a more holistic evaluation framework (Jacob et al., 2025; El Arab et al., 2025).

Adoption evaluation efforts should examine how AI tools align with existing clinical processes and identify potential benefits or disruptions. Examples of such methods include clinician usability testing, time-on-task analysis, and end-to-end workflow assessments. Human-centered evaluations should focus on the interaction between users and AI tools, addressing factors such as cognitive load, training and adoption dynamics, and trust between clinicians and AI systems.

Evaluation methodologies should account for the diverse societal, cultural, and economic variations across U.S. healthcare systems to ensure adaptability and inclusivity. By taking this multifaceted approach, stakeholders can better understand the practical value and equitable implementation of AI tools across clinical environments. This perspective aligns with the AI for IMPACTS evaluation model (Jacob et al., 2025), which emphasizes holistic, sociotechnical assessment across dimensions such as integration, monitoring, performance, acceptability, cost, safety, and impact.

Incentive Structures to promote AI Development and Optimization

To best support the development and optimization of these processes and to encourage the accuracy and reliability of AI tools, HHS should continue its tradition of publicly supported challenges with academia and industry. Regardless of the method used, HHS should continue to encourage innovation and exploration of AI tools related to accuracy, security, and end user satisfaction.

Data Strategy for Improved Interoperability

Another critical focus for supporting AI adoption and development in healthcare spaces is enhanced interoperability in healthcare data exchange. This serves as a critical enabler for expanding market opportunities, fueling research innovation, and accelerating the development of AI for clinical care. By harmonizing diverse data types including structured Electronic Health Record (EHR) data, imaging data (DICOM), genomics and omics data, and patient-generated data, healthcare systems can unlock richer AI insights and applications. For instance, structured EHR data builds the foundation for predictive tools to improve chronic care management, while imaging and genomics provide the basis for precision diagnostics and personalized medicine. Patient-generated health signals offer longitudinal, real-time monitoring for risk prediction services, and unstructured text enriched through NLP creates opportunities for enhanced clinical reasoning. These advances empower vendors to design broader AI solutions that reach larger markets, including underserved patient populations and specialized medical services.

Interoperability is further strengthened by standardized frameworks such as HL7's Fast Healthcare Interoperability Resources (FHIR) and Common Data Models (CDMs) like the Observational Medical Outcomes Partnership (OMOP). FHIR, established as the standard for healthcare data exchange through the 21st Century Cures Act, enables seamless communication across electronic

health records (EHRs) and digital tools using modular APIs (Office of the National Coordinator for Health IT [ONC], 2026). FHIR simplifies app integration—allowing developers to plug into EHR systems, fueling innovation in scalable healthcare platforms. Similarly, CDMs, like OMOP and Sentinel, harmonize data from diverse institutions to facilitate multi-site research and reproducible analytics for the FDA and NIH. By aligning clinical data standards for research, interoperability accelerates market entry for AI tools while encouraging collaboration across the healthcare ecosystem.

In addition to improving data access, benchmarking tools rooted in interoperability play a crucial role in accelerating the development and adoption of clinical AI systems. Tools like FHIR-AgentBench assess AI agents on realistic clinical tasks, test their ability to extract data, reason within HL7 FHIR structures, and choose appropriate next steps without "hallucinating" responses (Lee et al., 2025). By evaluating AI models on their ability to employ interoperable frameworks effectively, benchmarks ensure that vendors prioritize functionality and data integration, not just predictive accuracy.

Interoperability creates opportunities to fuel research by eliminating barriers to multi-institutional data sharing and collaboration. Unified frameworks like FHIR enable academic researchers, regulators, and industry stakeholders to analyze diverse datasets. Longitudinal and multimodal data across geographic and demographic settings fosters equity-focused outcomes research, while interoperable APIs make it easier for developers to refine tools for clinical trials and population health analytics. Research collaborations enabled by interoperable systems attract investment while ensuring that clinical AI pipelines remain evidence-based and scalable.

It would be valuable for HHS to **create a health care centered benchmark dataset** against which AI tools can be consistently evaluated. This dataset could consist of task specific use cases and be used to evaluate documentation assistance, triage, care coordination, and risk prevention scenarios that AI systems are likely to support in real world clinical environments. Benchmark tasks should reflect realistic clinical workflows, data availability at the point of care, and the longitudinal nature of patient management rather than isolated predictions.

The constructed dataset should be representative of the patient population and have set robustness and performance metrics. These metrics could include sensitivity, recall, precision, and Area Under the Curve (AUC). It could also evaluate robustness and sensitivity to noisy or missing data. The creation of the benchmark dataset would support reproducibility and reduce duplicative evaluation efforts across the whole health care system. **By treating benchmarking infrastructure as a public good, HHS would lower barriers to entry and increase trust in the role of AI systems in clinical care.**

From streamlining AI tool integration to enabling reproducible research on diverse data sets, interoperability not only improves the quality of healthcare data but also positions the industry for innovative, scalable solutions that address emerging challenges in clinical care. By harmonizing data and setting industry-wide benchmarks, healthcare systems can unlock the full potential of AI to improve patient outcomes and advance precision medicine.

AI Security and Governance

While it is important that AI tools are accurate, it is also important to ensure security, governance and patient privacy and safety when implementing AI in healthcare settings. AI tools used in clinical care which are considered non-medical devices (such as documentation assistants, summarization tools, etc.) may still influence clinical judgement, prioritization, and work processes in ways that create risks around patient safety, privacy, and accountability. These tools often operate as infrastructure within workflows, which can place them outside traditional regulatory categories. This creates risk in areas not well addressed by traditional compliance frameworks.

- **New legal and implementation issues introduced.** AI tools can introduce legal and implementation issues including influence on clinical decision-making, evolution in accuracy over time (through model updates or retraining, which can impact repeatability or unexplained differences in behavior), and probabilistic rather than deterministic operation. Existing governance and accountability structures typically assume functionality which does not change from one day to the next and assume deterministic outputs.
- **Liability and accountability ambiguity.** Non-device AI tools may shape decisions without clear assignment of responsibility among clinician, health system, and vendor. This ambiguity is amplified by automation bias and by AI outputs presented in persuasive or sycophantic natural language.
- **Indemnification and contracting gaps.** Many organizations lack leverage or technical capacity to negotiate appropriate vendor obligations related to model updates, monitoring, incident response, and performance drift.
- **Auditability and documentation limitations.** AI tools may not log model versions, inputs, outputs, or clinician interactions in a structured way, complicating quality assurance, root-cause analysis, and legal defensibility. AI tools are often ‘black-box’, challenging auditability requirements.
- **Privacy and secondary data use concerns.** Foundation model deployments raise uncertainty around PHI handling, data retention, and whether patient data may be used for training or fine-tuning, even when HIPAA compliance is asserted.
- **Security risks from expanded integration.** AI tools often require broad access to EHRs and clinical systems, increasing the attack surface and introducing new vulnerabilities through third-party services and model manipulation risks.
- **Human factors risk (cognitive load, alert fatigue, automation bias).** Even highly accurate tools can increase clinician burden or encourage over-reliance when outputs are verbose, confident, and timesaving. These risks are not captured by traditional performance metrics alone. For example, if AI is implemented poorly, it can create more clicks, more alerts leading to user fatigue and impeding adoption.

HHS is uniquely positioned to support consistent governance of non-device AI through coordinated action across federal agencies. HHS should prioritize:

1. **Baseline governance guidance** for AI tools used in clinical workflows including minimum expectations for monitoring, human oversight and escalation, and regulatory compliance. One approach could be to adapt existing models for risk management in healthcare to develop guidelines and standards for AI model approval processes.

2. **Setting standardized transparency and auditability expectations**, such as logging of model version, input sources, clinician interaction outcomes (accepted/edited/overridden), labeling if a record is AI generated, and confidence indicators (if applicable), to name just a few possible metrics.
3. **Evaluation and monitoring infrastructure**. Coordinating development of evaluation frameworks and benchmarking methods for non-device AI, including outcomes and human factors measures (such as cognitive load, review fatigue, automation bias).
4. **Privacy and security clarification** for appropriate use of protected health information in AI systems, such as acceptable architectures and protocols for cloud-hosted models, third-party data handling, and de-identification expectations for training or fine-tuning.
5. **Model procurement templates / recommended contracting clauses** addressing update notification, vendor responsibilities for drift, incident response, auditability, and data retention. One potential avenue would be to publish recommended contract clauses for common themes like transparency requirements, audit rights, disclosure of training data, and update notifications.

HHS can effectively support advancements in these areas through mechanisms such as grants, cooperative agreements, public-private partnerships, and prize competitions. Additionally, HHS could play a crucial role in reducing uncertainty and fostering the safe adoption of non-device AI by establishing governance expectations that are clear, transparent, and explicit. This would mitigate risks associated with fragmented accountability, which often arises from opaque contract language and complex workflow integrations.

Certification and Industry-Driven Testing of AI Tools

It follows naturally that if the AI tools themselves must be governed, so do the industries and practitioners developing, installing, and maintaining them. The Senate's overwhelming rejection (99-1) of proposed legislation for a 10-year moratorium on state and local AI regulations favoring centralized federal oversight highlights the challenges of decentralized regulatory influence on AI (U.S. Senate Committee on Commerce, Science, & Transportation, 2025). The EU has shown promising regulatory initiatives due to its ability to bring 27 countries together with one voice and its mandatory conformity assessments for high-risk medical AI. It also requires extensive documentation of data provenance, fairness safeguards, and human oversight. Through the EU's Artificial Intelligence Act (AIA) algorithmic integrity is embedded within regulatory compliance frameworks. The United States does not have such an overarching regulatory system. HHS has an opportunity to center itself as a leader in this space to promote a regulatory framework that supports the responsible, safe, and equitable use of AI.

HHS can further advance responsible AI adoption by enabling industry-driven testing through voluntary federal certification frameworks for third-party AI certifiers, rather than certifying AI tools directly. By publishing technical guidance, reference datasets, and evaluation frameworks, HHS can create a common foundation for independent testing and encourage transparency around performance, safety, and bias. The Office of the National Coordinator (ONC) Health IT Certification Program provides a strong precedent: ONC defines certification criteria and recognizes authorized certification bodies, while private entities conduct testing and certification of health IT products. This approach creates market pull, as hospitals, payers, and providers

preferentially adopt technologies with recognized credentials, and could similarly incentivize AI developers to participate in rigorous, independent evaluation.

Accreditation of organizations, programs, and systems. HHS can support private-sector accreditation by recognizing or endorsing independent accrediting bodies that assess organizational AI governance, model lifecycle management, and safety practices. This approach allows HHS to promote consistent standards for clinical AI while leveraging private expertise and avoiding duplicative oversight. A well-established precedent is CMS’s “deeming authority,” under which approved national accrediting organizations, such as The Joint Commission and DNV, are authorized to survey healthcare facilities for compliance with Medicare Conditions of Participation. Importantly, CMS retains oversight through validation surveys, demonstrating how federal agencies can rely on private accreditation while maintaining accountability and public trust. This model could be extended to accrediting AI governance programs and deployment environments in healthcare organizations.

Public-private partnerships to enable scalable testing ecosystems. Public-private partnerships (PPPs) allow HHS to create the conditions necessary for credible, scalable, industry-led testing without positioning the federal government as the tester of record. Through PPPs, HHS can co-fund pilots, sandboxes, and shared infrastructure, responsibly provide access to federal data assets, and convene stakeholders to align on evaluation priorities. The NIH Science and Technology Research Infrastructure for Discovery, Experimentation, and Sustainability (STRIDES) initiative illustrates this model by connecting NIH researchers and NIH-funded institutions to commercial cloud services at discounted rates (NIH, 2025). STRIDES demonstrates how HHS can reduce technical and financial barriers, enabling the private sector to build and test advanced biomedical and AI capabilities at scale.

Credentialing and workforce development for individuals. HHS can support private-sector credentialing and workforce development by defining high-level AI competency expectations, funding external credentialing initiatives, and encouraging healthcare organizations to recognize independently issued AI credentials. In this model, credentials are developed, issued, and maintained by professional societies, certifying bodies, or training organizations—not by HHS itself. HHS’s role is to align incentives and signal the importance of qualified AI use in clinical care. Medicare and Medicaid provider credentialing offers a clear precedent: CMS does not credential clinicians directly, but relies on private licensure, board certification, and employer-led credentialing processes to determine eligibility for participation in federal programs. A similar approach could be applied to AI-related clinical and oversight roles, ensuring safe and effective use while supporting a private credentialing ecosystem.

6.2 AI Adoption: Successes and Failures

There are many examples of AI being used or having been attempted in clinical settings. Reception of these tools are mixed due to variations in quality, functionality, rollout, user optimization, training and maintenance. For example, Legacy Health IT, including many EHR systems were designed around billing, not care. AI is now being layered upon that same architecture, so it naturally optimizes the wrong thing. Another example includes the CMS ACCESS models which lack software built directly into the EHR. Third party apps (Oracle, Epic, Meditech) connect via

CMS-required APIs instead, creating inefficiencies. It follows that AI will need its own platform for clinical care.

To date, AI tools in clinical care have demonstrated a stark contrast between their performance in administrative and diagnostic "closed loop" tasks versus complex bedside decision-making.

Where AI Has Met or Exceeded Expectations

- **Automated Clinical Documentation:** AI "scribes" (e.g., Nuance, DAX, Copilot) have significantly reduced administrative burnout. Organizations like Rush University Medical Center reported a 74% reduction in clinician burnout, while Northwestern Medicine saw doctors able to treat ~11 more patients monthly due to time savings (Dahdah, 2024; Siwicki, 2025). Several recent studies support these anecdotal reports, indicating significant reductions in task load and burnout along with reports that physicians could see additional patients because of time saved (Albrecht et al., 2024; Shah et al., 2025).
- **Imaging and Diagnostics:** AI has reached maturity in radiology and pathology. Tools often rival or exceed human accuracy in detecting colonic polyps, breast cancer in mammograms, and early-stage skin cancers. Jones et al. (2026) speaks to a significant finding in the use of AI and genomics data to assist in diagnosis of Pneumonia, a finding which could result in more than an 80% decrease in inappropriate antibiotics (Phan et al., 2025).
- **Operational Efficiency:** AI-driven scheduling at organizations like HCA Healthcare has yielded a 40% increase in productivity and a 60% enhancement in patient satisfaction by reducing wait times (Waligora et al, 2025).
- **Patient Triage and Virtual Assistance:** Chatbots and symptom-checkers integrated into patient portals have successfully cut unscheduled emergency visits while providing safe triage advice (Elechi et al., 2025).

Where AI Has Fallen Short

- **Real-World Predictive Accuracy:** While models often show "excellent" accuracy in controlled trials, they frequently fail at the bedside. A March 2025 study found that mortality-prediction models missed 66% of high-risk cases in real-world hospital settings (Pias et al., 2025).
- **Early Disease Detection (Sepsis):** Despite massive investment, many sepsis-detection models have underperformed. Research at the University of Michigan revealed a widely used model was wrong more often than if it was right, missing two-thirds of actual cases while triggering frequent false alarms (Lu et al., 2021).
- **Complex Clinical Reasoning:** Leading AI systems struggle when standard medical questions are modified to include "none of the above" or require managing incomplete information, with accuracy dropping by more than a third in some tests.
- **Scalability and Workflow Friction:** An estimated 80% to 95% of AI projects fail to scale beyond the pilot phase. Failure often occurs not due to technical flaws but because tools are not integrated into existing workflows, forcing clinicians to log into separate systems and adding "friction" rather than reducing it.

- **Bias and Cultural Nuance:** Many models exhibit calibration drift or perform poorly for non-white or low-income patient demographics because they were trained on limited, non-diverse datasets.
- **Failure to include patients in AI development:** Healthcare has struggled to include patients in AI development, often prioritizing technical, administrative, or efficiency goals over user-centered design, leading to tools that disrupt clinical workflows and fail to address actual patient needs. Key failures include opaque, non-inclusive development processes, lack of patient input on, and inattention to, potential biases.

The use of AI in medical practices nearly doubled, from 38% in 2023 to 66% in 2024, highlighting increased adoption of AI as an adjunct to patient care (American Medical Association, 2025). However, providers are most comfortable with AI tools related to administration, such as billing, transcription, discharge instructions, translation, research, and assistive diagnosis. A 2024 survey of the American Medical Association (AMA) revealed that doctors want AI tools that save time, improve accuracy, and strengthen patient care (AMA, 2025). In addition, they also want guarantees such as transparency, oversight, and ethical guardrails. Integration is a key consideration. In this survey, physicians emphasized the need for seamless EHR integration (84%), data privacy assurances (87%), and feedback channels (88%) to drive adoption (AMA, 2025). There is a recognized need to partner with AI not only for clinical input but for vendor liability when the AI tool gets it wrong.

Furthermore, with respect specifically towards the use of LLMs for treatment, rather than administrative purposes, research testing the use of LLMs for mental health treatment also raised concerns for both privacy, safety, and for therapeutic outcomes (Moore et al., 2025). These concerns included inappropriate chatbot responses to common and critical conditions as well as expression of stigma towards mental health conditions (Moore et al., 2025). For this reason, in most clinical settings, human interaction is still considered best practice for safe and adequate care.

While much of the discussion has been focused on the clinicians, administrators, vendors, and regulators of AI, **patient input into AI expectations is extremely underrepresented in research** (Moy et al, 2024; Nong & Ji, 2025). Studies, consisting mostly of smaller focus groups, have investigated patient perspectives on AI use in healthcare and generally surfaced the following priorities for AI:

1. That human oversight and decision making remains at the forefront of healthcare delivery.
 2. That patients be involved in the AI development as stakeholders.
 3. That the solutions be user friendly, easy to understand, and transparent in nature.
 4. That data privacy remains of utmost importance.
- (Foresman et al., 2025, Gundlack et al., 2025; Mikkelsen et al., 2023; Osnat, 2025)

Despite the concerns raised among patients regarding AI use in healthcare, it is also clear that many see the advantages of using these tools and hope they result in faster information delivery, reduced wait times, lower costs, and more personalized healthcare (Gundlack et al., 2025; Mikkelsen et al., 2023; Osnat, 2025). A 2023 study by Adus, Macklin, & Pinto explored patient perspectives on patient engagement with AI in healthcare and found that patients felt strongly about the importance of patient involvement in AI development as well as comprehensive AI education. Inclusion of diverse demographics in AI development was emphasized in this focus group study as vital to useful AI integration (Adus, Macklin, & Pinto, 2023). In summary, while

patients recognize the potential benefits of AI in healthcare, they emphasize the need for human oversight, transparent processes, and equitable representation in AI development. Prioritizing patient perspectives will be essential to ensure that AI solutions align with the needs and values of those they aim to serve.

7.0 Conclusions

Artificial Intelligence (AI) has the potential to transform healthcare by enhancing efficiency, reducing clinician burnout, improving diagnostic accuracy, and expanding access to underserved populations. However, for this transformation to manifest, critical barriers must be addressed. Trust by clinicians, patients, and administrators in the tools being used must be earned through transparent algorithms, pre- and post-deployment monitoring, bias mitigation, clear standards and regulatory requirement, and adoption of data governance frameworks for adoption, evaluation, and maintenance. A comprehensive, human-centered approach that emphasizes safety, accountability, equity, and long-term sustainability is essential to the successful integration of AI into clinical workflows.

This paper identifies actionable strategies across the healthcare landscape to enable responsible and scalable adoption of AI in healthcare. By transitioning to value-based payment models, improving data interoperability, fostering public-private collaborations, and implementing robust evaluation frameworks, AI can be integrated in ways that align with clinician trust, patient safety, and organizational priorities. Additionally, emphasizing human-in-the-loop design principles, transparency protocols, and dynamic post-deployment monitoring provides critical safeguards to mitigate risks such as automation bias, performance drift, and inequities in patient care as well as common issues with LLMs and hallucination.

The path forward must prioritize alignment across stakeholders, including clinicians, patients, vendors, and policymakers, to ensure that AI tools meet the diverse and evolving needs of the healthcare system. As AI continues to advance, the U.S. healthcare system has an opportunity to leverage its transformative capabilities while mitigating risks to achieve improved health outcomes at lower costs. Through strategic action, public-private collaboration, and innovative governance mechanisms, AI can drive solutions to longstanding challenges such as workforce shortages, systemic inefficiencies, and disparities in care.

With its extensive expertise in operational transformation and mission-first innovation, LMI stands ready to partner with HHS and other stakeholders to shape the future of healthcare.

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